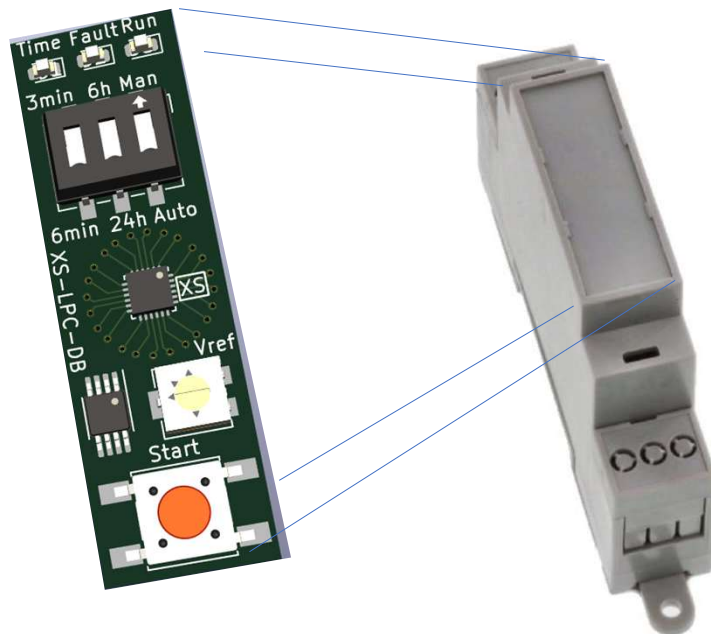




Purpose

- Deterministic current-sensing controller that replaces mechanical float switches with reliable automatic pumping.

Features

- Single CT input with 60 Hz CORDIC RMS engine for stable motor-current sensing
- 12-bit internal ADC for precise current-profile capture
- Front-panel threshold trim sets cutout margin below run current
- Current-cutout logic detects empty-tank, jam, or timeout
- Automatic periodic pumping (6 hr / 24 hr DIP-selectable cycle)
- Timed pump-out window (3 min / 6 min DIP-selectable max run)
- Manual pump-out button for immediate emptying
- Manual mode follows button directly for setup and testing
- Status LEDs: RUN (on-pump), FAULT (too long/too short/overcurrent), TIME (pending cycle)
- TIP122 open collector relay drive transistor
- Three terminal system interface (Pwr, Gnd, OC relay drive)
- Integrated PLL, oscillator, and POR — fully self-contained chip
- Always-works architecture — no firmware, no crashes, no jitter
- DIN-module integration with upper logic/HMI board and lower power/relay board
- Development board (DB) with emulator header

Typical Applications

- Automatic water-level control for residential pump systems Reliable float-switch replacement using learned motor-current behavior.
- Current-based pump control in compact commercial equipment Solid-state activation and cutout for small tanks and reservoirs.
- Embedded pump logic for OEM assemblies Deterministic current-profile switching without mechanical floats.